
Daniel Boros

Senior Software Engineer — Researcher — Financial Mathematics Specialist
Hungary

Contact Information

Email: dancixx@gmail.com

LinkedIn: [linkedin.com/in/daniel-boros-b86a5373](https://www.linkedin.com/in/daniel-boros-b86a5373)

Summary

With a strong foundation in mathematics and a deep focus on high-performance systems, I have built a career at the intersection of software engineering, quantitative research, and artificial intelligence. I am currently a Senior Rust Engineer at [Qdrant](#), where I work as a core engineer on scalable cloud inference and core product development. In parallel, I am a part-time Core Rust Engineer and Quant Lead Engineer at AlgoFusion, focusing on the design and implementation of trading engine infrastructure.

Previously, I worked as a Senior Software Engineer at [Kadasolutions](#), developing backend, AI, and data-processing systems in Rust. I have also been a Researcher at the [AI Research Group at Eötvös Loránd University, Institute of Mathematics](#), where my work has focused on deep learning, parameter estimation of fractional stochastic processes, and the generation of fractional stochastic processes, including maintaining `stochastic-rs`.

My expertise includes Rust, Python, TypeScript, React.js, cloud infrastructure, AI systems, and quantitative software engineering. I am proficient in English and German. I hold a Master's degree in Actuarial and Financial Mathematics and a Bachelor's degree in Civil Engineering.

Experience

Senior Rust Engineer

October 2025 - Present

Qdrant, Germany

- Core engineer at Qdrant; focusing on scalable cloud inference and core product.

Core Rust Engineer

Aug 2025 - Present

AlgoFusion, Switzerland, Part-Time

- Quant lead engineer
- Core Rust engineer; focusing on the design and implementation of the trading engine.

Senior Software Engineer

December 2022 - October 2025

Kadasolutions, Hungary

- Backend development in Rust (Tokio, Axum, Tower, Actix, Rumqtt, Rayon)
- AI development in Rust (Candle, OpenCV-Rust, Onnx, Tch-rs, Ndarrray, Similari-rs)
- Data preprocessing with GeoRust and parallelization with Rayon

Researcher

June 2021 - December 2025

AI Research Group, Eötvös Loránd University, Hungary

- AI development in Rust, maintaining `stochastic-rs`
- Researching deep learning and parameter estimation of fractional processes

Chief Technology Officer

September 2018 - Present

GymPiper Ltd., Hungary

Led the software development department for mobile and web apps. Managed backend systems using Rust, TypeScript, and Google Cloud Platform. Worked with Hasura, GraphQL, React, and Python (Keras/TensorFlow).

Principal CFO

September 2012 - August 2018

GetPro Kft., Hungary

Responsible for financial stability and accounting. Taught advanced mathematics and mechanics to engineering students.

Recent Projects

- **Co-Owner of Rust-DD Open-Source Rust Community**

I founded and manage the [Rust-DD](#) open-source community, aimed at fostering development in the Rust ecosystem. We also maintain a website at [rust-dd.com](#).

- **Creator and Maintainer of Stochastic-RS**

[Stochastic-RS](#) is a Rust library I created and continue to maintain. It focuses on generating fractional stochastic processes to provide synthetic data for machine learning models.

- **Real-Time Stream Server for Drone Streams**

Developed a real-time stream server in Rust for managing drone video streams.

Tech Stack: Tokio, Axum, Hasura, GraphQL, PostgreSQL

- **Object Detection using Local LLM and YOLO/Detectron2**

Currently working on an object detection project leveraging local large language models (LLM) combined with YOLO and Detectron2 for image processing on real-time drone streams.

Tech Stack: Tokio, Axum, Hasura, GraphQL, PostgreSQL, Candle-RS, Mistral-RS, OpenCV-Rust, Numpy, Similarity-RS

- **Big Data Processing in Rust**

Managing a big data processing system that handles over 50TB of data daily.

Tech Stack: Tokio, Polars, Rayon

- **IoT MQTT Server**

Building an IoT MQTT server for high-performance data streaming.

Tech Stack: Tokio, Axum, Hasura, PostgreSQL, GraphQL, Rumqtt, EMQX

- **Creator of NOEX Fitness**

Founded [NOEX Fitness](#), a fitness platform integrating both mobile and web applications for a comprehensive fitness solution.

Tech Stack: React-Native, Typescript, Hasura, GraphQL, PostgreSQL, Tokio, Axum

Education

PhD, Computational and Applied Mathematics

September 2022 - Present

Eötvös Loránd University, Hungary

Master's Degree, Actuarial and Financial Mathematics

September 2019 - January 2022

Corvinus University of Budapest, Hungary

Bachelor's Degree, Civil Engineering

2009 - 2017

Budapest University of Technology and Economics, Hungary

Skills

Programming Languages:

Proficient in a range of languages including **Rust**, **TypeScript**, **R**, **Python**, utilized for diverse applications ranging from system-level programming to web development and data analysis.

Frontend Frameworks:

Experienced in building user interfaces using **React** and **React-Native** with TypeScript, as well as leveraging Rust and Wasm-based frameworks like **Leptos**, **Dioxus**, and **Tauri**.

Backend Frameworks:

Specialized in developing scalable and high-performance backend systems using **Tokio**, **Axum** (preferred), **Actix**, and **Tower** in Rust. Additionally experienced in **Node.js** and **Hono.js** with TypeScript for lightweight and performant web services.

AI and Machine Learning Development:

Experienced in AI and machine learning tools, leveraging **TensorFlow**, **Keras**, **OpenCV-Rust**, **Tch-rs**, **Candle-rs** from Hugging Face, and **Mistral.rs** for building deep learning models, computer vision applications, and complex data processing pipelines.

Data Processing and Analysis:

Expert in handling large datasets and performing efficient data analysis using tools like **GeoRust**, **Rayon**, **Polars**, **Ndarray**, **Pandas**, and machine learning libraries such as **Scikit-learn (Sklearn)**.

Publications

- Dániel Boros et al. “A Fractional Process with Jumps for Modeling Karstic Spring Discharge Data”. In: *Mathematics* 13.18 (2025). ISSN: 2227-7390. DOI: [10.3390/math13182928](https://doi.org/10.3390/math13182928). URL: <https://www.mdpi.com/2227-7390/13/18/2928>
- Dániel Boros et al. “Deep learning the Hurst parameter of linear fractional processes and assessing its reliability”. In: *Quality and Reliability Engineering International* (2024). DOI: [10.1002/qre.3641](https://doi.org/10.1002/qre.3641). URL: <https://doi.org/10.1002/qre.3641>
- Bálint Csanády et al. *Parameter Estimation of Long Memory Stochastic Processes with Deep Neural Networks*. 2024. arXiv: [2410.03776](https://arxiv.org/abs/2410.03776) [cs.LG]. URL: <https://arxiv.org/abs/2410.03776>

Honors & Awards

- **Certificate of Best Presentation Award:** Honored to receive this award at the XVII. International Conference on Mathematical Finance and Economics (ICMFE 2023) in New York. The recognition was for our presentation on “Fast Estimation of Fractional Process Parameters in Rough Financial Models Using Artificial Intelligence.” It’s a privilege to contribute to the field alongside my esteemed co-authors, and this accolade underscores our commitment to innovative research at the nexus of finance, mathematics, and AI.

Certificates

- **Beginning C++ Programming - From Beginner to Beyond** (Issued Feb 2022)
[Credential ID UC-b313cd64-a3ad-4206-bd54-c21431f3228e](#)
- **Understanding TypeScript 2022 Edition** (Issued Feb 2022)
[Credential ID UC-7a1277b8-b425-4e92-9b3f-ec1682f3f8e8](#)
- **Google Cloud Functions with Python** (Issued Apr 2021)
[Credential ID UC-7d985625-99cf-4e8b-982f-382cbda7fbdf](#)
- **Learn Python Programming Masterclass** (Issued Apr 2021)
[Credential ID UC-3d63116b-ba94-45a5-a6e5-41e2147539b4](#)
- **React, Redux and Firestore** (Issued Apr 2020)
[Credential ID UC-c40c5ba0-9d8b-4985-8628-a9398b7f1abd](#)
- **GraphQL with React: The Complete Developer’s Guide** (Issued Jan 2020)
[Credential ID UC-NMH5C1SO](#)
- **The Complete React Developer Course (w/ Hooks and Redux)** (Issued Nov 2019)
[Credential ID UC-VFV9Y67M](#)
- **The Modern GraphQL Bootcamp (with Node.js and Apollo)** (Issued Nov 2019)
[Credential ID UC-OTVEMYWF](#)
- **The Modern JavaScript Bootcamp (2019)** (Issued Nov 2019)
[Credential ID UC-BKIXPZDA](#)
- **The Complete Node.js Developer** (Issued Oct 2019)
[Credential ID UC-9IWDSZOD](#)